Time–Optimal $T_{\text{on}}/T_{\text{off}}$ Computation

STM32G474 Voltage–Mode Synchronous Buck — Large–Signal Load–Step Response

G. Qian

September 25, 2026

Abstract

This note documents exactly how the firmware computes the single ON–OFF switching times $T_{\text{on}}, T_{\text{off}}$ that are applied for one stretched HRTIM period on a detected load step–up. It covers (i) the second–order model and the two terminal equations, (ii) their reduction to a single quadratic (one square root) as implemented in `buck_optimal_onoff()`, (iii) root selection and clamps, (iv) the ISR closed form that replaced it and its pipeline (peak–current limit, time clamps, tick quantization), and (v) the Jacobian timing adaptation and the post–pulse reseed. Symbols map one–to–one to the C source.

1 Plant model and shifted state

The synchronous buck averaged over a switching action is a second–order LC plant with a lumped series resistance r (inductor DCR + FET/path):

$$L\,\frac{di_L}{dt} = v_{\text{sw}} - V_{\text{out}} - r\,i_L, \qquad C\,\frac{dV_{\text{out}}}{dt} = i_L - i_{\text{load}}, \tag{1}$$

with $v_{\text{sw}} = V_{\text{in}}$ while the high side is ON and $v_{\text{sw}} = 0$ while it is OFF. Define the ramp rates

$$\theta = -\frac{r}{L}, \qquad m_1 = \frac{V_{\text{in}} - V_{\text{ref}}}{L}, \qquad m_2 = \frac{V_{\text{ref}}}{L}, \qquad m_\Sigma = m_1 + m_2 = \frac{V_{\text{in}}}{L}. \tag{2}$$

After a load step the controller works in the *shifted state* measured at the step (all in SI units), taken from the Padé current/load estimator:

$$z_i = i_L - \hat{I}_{\text{load,new}}, \qquad z_v = V_{\text{out}} - V_{\text{ref}}, \qquad \hat{I} = \hat{I}_{\text{load,new}}. \tag{3}$$

In the code (`buck_optimal_service`), i_L = `cap_iavg_q20`, $\hat{I}$ = `cap_iload_q20` (both Q20 A), and V_{out} = `cap_adc`×`UV_PER_COUNT`. The goal is to drive $(z_i, z_v) \to (0, 0)$ in *one* ON–OFF action of durations $T_{\text{on}}, T_{\text{off}}$.

2 Terminal equations

Propagating the shifted state through the ON interval of length $x = T_{\text{on}}$ then the OFF interval of length $y = T_{\text{off}}$ with the second–order bilinear (Padé) state–transition and requiring the final shifted state to vanish yields two coupled equations (TPEL Sec. IV, eqs. 35–36; verified against a direct Newton solve to $\sim 10^{-16}$ in `Tests/test_optimal.c`):

$$0 = z_i + p\,x + q\,y + g\,x\,y, \tag{4}$$

$$0 = z_v + \tfrac{z_i}{2C}\,(x + y) + b\,x\,y, \tag{5}$$

with coefficients

$$p = \tfrac{\theta}{2} z_i - \tfrac{z_v}{2L} + m_1 - \theta \hat{I}, \qquad q = \tfrac{\theta}{2} z_i - \tfrac{z_v}{2L} - m_2 - \theta \hat{I}, \tag{6}$$

$$g = \tfrac{1}{4}\left(\theta^2 - \tfrac{1}{LC}\right) z_i - \tfrac{\theta z_v}{4L} + \tfrac{\theta m_\Sigma}{2}, \qquad b = \tfrac{\theta z_i}{4C} - \tfrac{z_v}{4LC} + \tfrac{m_\Sigma}{2C}. \tag{7}$$

A useful identity used below is

$$p - q = m_1 + m_2 = m_\Sigma = \frac{V_{\text{in}}}{L} \implies \frac{1}{p-q} = \frac{L}{V_{\text{in}}}. \tag{8}$$

3 Reduction to one quadratic (single square root)

Let $s = x + y = T_{\text{on}} + T_{\text{off}}$ and $b_1 = \dfrac{z_i}{2C}$.

Step 1 — eliminate the bilinear term. From (5), $b\,xy = -(z_v + b_1 s)$, i.e.

$$x\,y = -\frac{z_v + b_1 s}{b}. \tag{9}$$

Step 2 — make T_{on} affine in s. Substitute $y = s - x$ and the product (9) into (4). The bilinear term $g\,xy$ becomes $-g\,(z_v + b_1 s)/b$, leaving a *linear* equation in x:

$$z_i + q\,s + (p - q)\,x - \frac{g}{b}\,(z_v + b_1 s) = 0. \tag{10}$$

Solving for x and using (8),

$$\boxed{x = T_{\text{on}} = a_0 + a_1 s, \qquad a_0 = -\frac{L}{V_{\text{in}}}\left(z_i - \frac{g\,z_v}{b}\right), \quad a_1 = -\frac{L}{V_{\text{in}}}\left(q - \frac{g\,b_1}{b}\right).} \tag{11}$$

Step 3 — one quadratic in s. With $x = a_0 + a_1 s$ and $y = s - x = (1 - a_1)s - a_0$, the product $xy = a_1(1 - a_1)s^2 + a_0(1 - 2a_1)s - a_0^2$. Inserting into (5) gives

$$A\,s^2 + B\,s + C_c = 0, \qquad \begin{aligned} A &= b\,a_1(1 - a_1), \\ B &= b\,a_0(1 - 2a_1) + b_1, \\ C_c &= z_v - b\,a_0^2, \end{aligned} \tag{12}$$

whose solution requires a *single* square root:

$$s = \frac{-B \pm \sqrt{B^2 - 4A\,C_c}}{2A}, \qquad T_{\text{on}} = a_0 + a_1 s, \qquad T_{\text{off}} = s - T_{\text{on}}. \tag{13}$$

The root (+ or −) is chosen as the physical one giving $T_{\text{on}} > 0$ and $T_{\text{off}} > 0$. If the discriminant is negative, or $A = b = 0$, no valid single–shot solution exists and no pulse is issued.

Note. The paper's boxed closed form does not satisfy (4)–(5); (11)–(13) is the re–derived form actually implemented, and it matches a Newton solve of (4)–(5) to floating–point precision.

4 As implemented: `buck_optimal_onoff()`

```
q = th*zi/2 - zv/(2L) - m2 - th*Ihat;
g = 0.25*(th*th - 1/(L*C))*zi - th*zv/(4L) + th*(m1+m2)/2;
b = th*zi/(4C) - zv/(4L*C) + (m1+m2)/(2C); b1 = zi/(2C);
a0 = -(L/Vin)*(zi - g*zv/b); a1 = -(L/Vin)*(q - g*b1/b);
A = b*a1*(1-a1); B = b*a0*(1-2*a1) + b1; Cc = zv - b*a0*a0;
disc = B*B - 4*A*Cc; // < 0 -> no solution
s = ((+/-)sqrt(disc) - B)/(2A); // pick root with Ton>0, Toff>0
Ton = a0 + a1*s; Toff = s - Ton; // then clamp to [tmin, *_max]
```

This float form is now the *host reference* (`Tests/test_optimal.c`). It ran in thread mode until September 2026, when the bench showed the pulse arriving $\sim$100 μs after the load edge: with the armed ISR at $\sim$244 of 280 clocks the thread gets 15–35 clocks per period, and this solve needs $\sim$50 periods of that. The firmware now solves in the ISR.

5 ISR closed form (September 2026): two square roots, one cycle

For a load step–up write $a = i_0 - I_{\rm eq} < 0$ (i_0 the valley at the pulse start, $I_{\rm eq} = \hat{I} + b$ the equilibrium current), $\Delta V = V_{\rm ref} - v_{\rm out} > 0$, $m_1 = (V_{\rm in} - V_{\rm ref})/L$. The terminal state is the *periodic steady state* of the PWM converter at $I_{\rm eq}$: the valley $I_{\rm eq} - h$, $h = \frac{1}{2}(V_{\rm in} - V_{\rm ref})(V_{\rm ref}/V_{\rm in})T_{\rm sw}/L$ (half the ripple, 2.45 A here), so that the next period is an ordinary one and the first sample after the pulse sits at the same point of the ripple as every steady–state sample. The fall from $I_{\rm eq} + x$ down to $I_{\rm eq} - h$ sees the series drop,

$$m_2' = \frac{V_{\rm ref} + r\, i_2}{L}, \qquad i_2 = I_{\rm eq} + \frac{x - h}{2}, \tag{14}$$

($r =$ `g_buck_opt_r_off_mohm`, a live tunable defaulting to the fitted `BUCK_R_MOHM`). The pulse delivers, above $I_{\rm eq}$, the ON triangle $(x^2 - a^2)/(2m_1)$ plus the OFF area $(x^2 - h^2)/(2m_2')$; equating the sum to the deficit $C\,\Delta V$ gives, with $m_1 + m_2' \simeq V_{\rm in}/L$,

$$x^2 = \frac{V_{\rm ref} + r i_2}{V_{\rm in}} \Big[\frac{2C\,(V_{\rm in} - V_{\rm ref})}{L}\,\Delta V + a^2 + \frac{(V_{\rm in} - V_{\rm ref})\,h^2}{V_{\rm ref} + r i_2} \Big], \qquad T_{\rm on} = \frac{x - a}{m_1}, \qquad T_{\rm off} = \frac{(x + h)\,L}{V_{\rm ref} + r i_2}. \tag{15}$$

The ideal–fall root x_0 ($r = 0$, $h = 0$) gives i_2, a second root gives x. On the plant the model assumes (series drop on both ramps, trapezoidal average) the pulse lands within 1.7 mV and 2 mA of ($V_{\rm ref}$, $I_{\rm eq} - h$) (`Tests/test_solve.c`). The former ideal fall $T_{\rm off} = xL/V_{\rm ref}$ was up to 2.4$\times$ too long at 10 A with $r = 114$ mΩ: on the bench every pulse swung the current far below the load and the adaptation ran to its bounds. Applied as in Section 7: the excursion is capped so that $I_{\rm new} + x_{\rm app} \le I_{\rm pk}^{\rm max}$ (`sat`), the rise above i_0 is scaled by $\gamma_{\rm on}$ and $T_{\rm on}$ is obtained from the resulting peak by the *same* trapezoidal inversion the current loop uses every cycle, $T_{\rm on} = L\,(I_{\rm pk} - i_L)/(V_L - r(i_L + I_{\rm pk})/2)$ (`buck_cm_ton_raw`, one bounded UDIV), so the pulse peak lands where the estimator predicts it, and $T_{\rm off} = \gamma_{\rm off}\, x_{\rm app} L/V_{\rm ref}$ (Section 6); then the time clamps and the stretched–period floor of (c)–(d). The terminal current reported to the reseed is the applied peak minus what the final off–time removes, clamps and floor included.

Fixed point. Currents in Q8 A inside the solve ($a^2 \le (32\,{\rm A})^2$ fits Q16 A^2), ΔV clamped to 2047 counts so the first product stays below 2^{31}, the factors $V/V_{\rm in}$ applied as Q32 high–word multiplies (`UMULL`), the h^2 term and $T_{\rm off}$ by bounded `UDIV`s. The roots are `buck_isqrt32`: normalise

$v \to v\,2^{2e} \in [2^{30}, 2^{32})$ with `CLZ`, read $\sqrt{i\,2^{24}}$ and its slope ($i = 64..255$, one 32–bit word each, 192 entries filled at start–up) and interpolate linearly, then shift back by e: ∼18 clocks, within 1 LSB of $\lfloor\sqrt{v}\rfloor$ over the 32–bit range. The coefficients that depend on V_{ref} and k are converted once by the thread (`buck_so_coef_from_float`, struct `g_buck.so`).

Timing (measured on the NUCLEO). The ISR is entered 134–142 clocks into the switching period (ADC trigger at 600 ticks, trigger resync, 12.5-cycle conversion at 56 MHz, EOC resync, exception entry), and the HRTIM preload window closes at 264 clocks; the solve takes 172 clocks on silicon. The pulse is therefore pipelined: trigger sample in period n (end of the hot path; it clears the hot path's `running` flag), solve in $n+1$ through the existing cold entry (no HRTIM write, no deadline; the solve is done for the state at the $n+3$ rollover: valley two periods ahead with the frozen duty, droop plus $(I_{\text{new}} - i_{\text{avg}})\,2T/C$), `PER`/`CMP1` written first thing in $n+2$ (measured 152–198 clocks into the period, 66–112 clocks before the deadline), stretched period $n+3$. Three switching periods (5 µs) from the trigger sample, against ∼53 with the thread solve. The every–cycle path is unchanged (244 clocks, static count).

6 Dual correction on the measured terminal state (manuscript Sec. V)

The applied pulse is $\mathbf{t}_a = \boldsymbol{\Gamma}\,\mathbf{t}^\star$, $\boldsymbol{\Gamma} = \text{diag}(\gamma_{\text{on}}, \gamma_{\text{off}})$ (eq. 40 of the manuscript): γ_{on} scales the physical on–time, i.e. the rise above the pulse–start current i_0, γ_{off} the off–time. The closed form (Section 5) aims at the periodic steady state ($I_{\text{eq}} - h$, V_{ref}), $I_{\text{eq}} = \hat{I} + b$ the load measured one period after the edge plus a learned correction b.

Since the inductor current is sensed, the terminal state of the pulse is *observed*, at the first sample after the stretched period (600 ticks into the first normal period, before the PI has changed anything):

$$z_{v,f} = v_s - V_{\text{ref}} - R_{\text{ESR}}\,(i_s - I_{\text{settled}}), \tag{16}$$

$$z_{i,f} = (i_s - m_1 t_s) - (I_{\text{eq}} - h), \tag{17}$$

where v_s, i_s are the sampled voltage and current, $m_1 t_s$ the rise from the valley to the sample instant (0.96 A at 61 ns), and $I_{\text{eq}} - h$ the target valley. The equilibrium itself is checked when the linear–controller transition ends (192 cycles, Section 6):

$$e_b = I_{\text{settled}} - I_{\text{eq}}, \tag{18}$$

I_{settled} the low–passed eq. 7 estimate at that moment (±0.15 A, against ±0.5 A for the former one–period window). Three residuals, three parameters, one thread event per pulse. With the terminal map from the pulse start,

$$z_{i,f} = z_{i0} + m_1 T_{\text{on}} - m_2 T_{\text{off}}, \tag{19}$$

$$z_{v,f} = z_{v0} + \tfrac{1}{C}\big[z_{i0}(T_{\text{on}} + T_{\text{off}}) + \tfrac{m_1}{2} T_{\text{on}}^2 + m_1 T_{\text{on}} T_{\text{off}} - \tfrac{m_2}{2} T_{\text{off}}^2\big], \tag{20}$$

$m_2 = (V_{\text{ref}} + r\,i_2)/L$ the fall slope of this pulse (i_2 the mean current of the fall), the Jacobian at the applied times is

$$\mathbf{J} = \begin{bmatrix} m_1 & -m_2 \\ \dfrac{z_{i0} + m_1(T_{\text{on}} + T_{\text{off}})}{C} & \dfrac{z_{i0} + m_1 T_{\text{on}} - m_2 T_{\text{off}}}{C} \end{bmatrix}, \tag{21}$$

and the *timing* correction is the manuscript's Newton step (eqs. 43–48) on the full measured residual,

$$\boldsymbol{\gamma}^{+} = \Pi_{\Gamma}\left[\boldsymbol{\gamma} - \mu\,\mathrm{diag}(\mathbf{t}^{\star})^{-1}\mathbf{J}^{-1}\begin{bmatrix} z_{i,f} \\ z_{v,f} \end{bmatrix}\right], \qquad \mu = 0.25, \quad \Gamma = [0.5,\, 2.0], \tag{22}$$

with at most 0.25 of change per pulse and $\det \mathbf{J}$ floored at $m_2 \cdot 0.5\,\mathrm{A}/C$. A clamped on–time (peak limit or $T_{\mathrm{on}}^{\max}$) cannot move the delivered charge: T_{off} alone then takes $\Delta T_{\mathrm{off}} = \mu\, z_{i,f}/m_2$ (the voltage residual is unreachable). The *equilibrium* correction is

$$b^{+} = b + \mu_b\, e_b, \qquad \mu_b = 0.5, \quad |b| \le 3\,\mathrm{A}. \tag{23}$$

The residuals are separable: a wrong equilibrium moves e_b and not the terminal residuals (the pulse still lands on its target), a wrong L, C or r moves the terminal residuals. *Validity:* $z_{i,f}$, $z_{v,f}$, e_b are clipped to $\pm 2\,\mathrm{A}$ / ± 40 counts before use, and a residual beyond $\pm 6\,\mathrm{A}$ / ± 200 counts, or a trigger on a droop beyond 200 counts, marks the record as corrupted: that pulse teaches nothing. (On the bench, 25 September 2026, the voltage sample read zero counts; the previous clip–only rule fired a pulse on every unmasked cycle and ran both learners to their bounds.) Host test (`Tests/test_solve.c`, the learners fed with the plant's actual terminal current and true load): a matched plant leaves $\gamma = 1$, $b = 0$; a 0.6 A load bias is found by b with γ untouched; L, C and r mismatches are absorbed within 21 pulses (the contraction per pulse is μ times the plant–to–model fall–slope ratio); a peak–clamped pulse has its current residual nulled by T_{off} alone.

Post–pulse operating point (reseed). With the current measured the voltage loop's reference is real amperes: after the pulse it is reseeded to I_{eq} itself and the current loop to the duty that holds it, $d_{\mathrm{trig}} + rT_{\mathrm{sw}}(I_{\mathrm{eq}} - \bar{i}_{\mathrm{trig}})/V_{\mathrm{in}}$ (the fitted r is the DC duty–to–current relation), both precomputed in the write cycle. The trigger is then masked for 192 cycles (320 μs, the linear–controller transition of manuscript Sec. VI) while the PI owns the recovery; when the mask ends the settled load closes the transient's record (e_b above) and the thread runs the dual correction. Without that mask a wrong reseed rings the loop, the ring's undershoot re–triggers and the loop limit–cycles (simulation: 19 pulses and a 1.2 V overshoot on a 1→6 A step with a 15 mΩ plant). Each transient writes one of two alternating records (trigger sample, solve inputs and outputs, reseed, terminal sample, settled load): no field the thread reads is ever live.

Measured current (average current mode). Since the fourth revision the inductor current is sensed (1 mΩ × 40 on A1, converted by ADC2 at the same instant as the voltage): the inner loop is an average–current PI, the load estimate is eq. 7 of the manuscript on the measured period average ($\bar{i}_L = i_{\mathrm{sample}} + (V_{\mathrm{in}} - V_{\mathrm{ref}})/L\,(T_{\mathrm{on}}/2 - t_s)$, t_s the sample's position on the on–ramp), i_0 is the measured valley extrapolated two periods with the measured drift and $\hat{I}$ the load measured over the period after the edge. The Padé recursion no longer runs in the ISR.

Capacitor ESR. The ADC sample is $v_C + R_{\mathrm{ESR}}\,(i_L - I_{\mathrm{load}})$ at the sample instant (on the bench: a +80–100 mV spike during the pulse at ∼7.6 A of capacitor current, i.e. 10–15 mΩ). Two corrections use the *measured* sample current: the solve refills $\Delta V = V_{\mathrm{ref}} - v_s + R_{\mathrm{ESR}}\,(i_s - \hat{I})$ rather than the raw error (during a droop the inductor lags the load, so the sample sits below the capacitor by that much: 30 counts for a 2 A lag at 12 mΩ, more than half the droop of a 3 A step), and the terminal–voltage residual is corrected against the settled load as above. R_{ESR} is a live tunable (`g_buck_opt_esr_mohm`) measured on the scope as the spike during the pulse divided by $I_{\mathrm{pk}} - I_{\mathrm{load}}$; the former in–loop identification read 60–75 % of the truth and is gone.

7 Applied $T_{\text{on}}/T_{\text{off}}$: gain, peak–current limit, clamps

The solver output $(T^{\star}_{\text{on}}, T^{\star}_{\text{off}})$ is post–processed in `buck_optimal_service()` before being written to the HRTIM:

(a) Timing gains

The excursion x of the closed form is capped at the peak limit ($x_{\text{app}} \le I_{\text{pk}}^{\max} - I_{\text{new}}$, flag `sat`); the applied rise above i_0 is $\gamma_{\text{on}}(I_{\text{new}} + x_{\text{app}} - i_0)$, re–capped at the peak limit, and the on–time is its trapezoidal inversion,

$$T'_{\text{on}} = \frac{L\,(I_{\text{pk,a}} - i_0)}{V_L - r\,(i_0 + I_{\text{pk,a}})/2}, \qquad T'_{\text{off}} = \gamma_{\text{off}}\,\frac{x_{\text{app}}}{m_2}, \tag{24}$$

with $\gamma_{\text{on}}, \gamma_{\text{off}}$ from Section 6 (1 at boot). The former learned gain k (excursion scaling with an empirical secant on the first post–pulse sample) is gone: its residual depended on the trigger timing, not on k.

(b) Peak–current–mode limit (control, not protection)

The on–time is capped so the predicted peak inductor current stays at a configured limit $I_{\text{pk,max}}$ (`BUCK_OPT_IPK_MAX_A`). With the estimator's inductor current i_L at the step as the ramp start,

$$I_{\text{peak}} = i_L + \frac{V_{\text{in}} - V_{\text{out}}}{L}\,T_{\text{on}} \le I_{\text{pk,max}} \implies T_{\text{on}} \le T_{\text{on}}^{I_{\text{pk,max}}} = \frac{(I_{\text{pk,max}} - i_L)\,L}{V_{\text{in}} - V_{\text{out}}}. \tag{25}$$

This is the normally binding limit; it only shortens T_{on} and never trips a fault (over–current *protection* is external to the loop).

(c) Time clamp and quantization

Finally

$$T_{\text{on}}^{\text{app}} = \min\!\big(T'_{\text{on}},\, T_{\text{on}}^{I_{\text{pk,max}}},\, T_{\text{on}}^{\max}\big),\ \ge t_{\min}, \qquad T_{\text{off}}^{\text{app}} = \max\!\big(T'_{\text{off}},\, t_{\min}\big), \tag{26}$$

where $T_{\text{on}}^{\max} < T_{\text{sw}}$ is required so that the compare write $\texttt{CMP1} = T_{\text{on}}$ is a valid compare against the (still active) one–period register. If either the peak or time clamp binds, a saturation flag excludes γ_{on} from the adaptation of that pulse. The seconds values are quantized to HRTIM ticks,

$$n_{\text{on}} = \Big\lfloor T_{\text{on}}^{\text{app}}/T_{\text{tick}} + \tfrac{1}{2} \Big\rfloor, \qquad n_{\text{off}} = \Big\lfloor T_{\text{off}}^{\text{app}}/T_{\text{tick}} + \tfrac{1}{2} \Big\rfloor, \tag{27}$$

and applied for one stretched period as $\texttt{PER} = n_{\text{on}} + n_{\text{off}}$, $\texttt{CMP1} = n_{\text{on}}$ (high side ON for n_{on}, OFF for the remainder).

(d) Stretched–period floor

The ISR that restores the normal PWM runs at a *fixed* point of the stretched period (ADC trigger at `CMP3` plus conversion and interrupt entry, roughly 4.5–5.3×10^3 ticks in). A stretched period shorter than that would end before the restore is written, the short pulse would repeat and the deadline check would trip. The firmware therefore enforces

$$n_{\text{on}} + n_{\text{off}} \ \ge\ N_{\min} = \texttt{BUCK_OPT_PER_MIN_TICKS} = \texttt{BUCK_HRTIM_PERIOD}, \qquad n_{\text{off}} \leftarrow N_{\min} - n_{\text{on}} \text{ if needed}, \tag{28}$$

i.e. the large–signal period is never shorter than a normal one, so every ISR timing assumption of a normal cycle holds inside it. With $k = 1$ the solver's $T^{\star}_{\text{on}} + T^{\star}_{\text{off}}$ is 2–4 μs for realistic captured states and the floor does not bind; it binds only when γ_{off} has shrunk T_{off}.

(e) Trigger

The state–anchored trigger that starts a transient acts on the output–capacitor state, both inputs in ADC counts: $z_v = v_s - V_{\text{ref}}$ and $d_v = v_s(n) - v_s(n-1)$, the one–period change ($C\,dv/dt$ is the capacitor current $i_L - I_{\text{load}}$: the load step itself, before the current loop has moved the inductor; 1 A is 8 counts per period). With the inductor current regulated, the former current–displacement axis only reported the current loop. Fire when $S = (z_v/V_a)^2 + (d_v/dV_a)^2 + 2\rho\,(z_v/V_a)(d_v/dV_a) > 1$ with $z_v < 0$ and $d_v \le 0$ (below the reference and not recovering: the only signature the step–up solver can act on). S is evaluated inside the ISR in integer arithmetic by completing the square, $S = u^2 + w^2$ with $u = n_v + \rho n_d$, $w = \sqrt{1-\rho^2}\,n_d$ ($n_v = z_v/V_a$, $n_d = d_v/dV_a$ in Q14, compared against 2^{28}), only for samples that pass the two sign tests. V_a (25 counts), dV_a (16 counts per period), ρ remain run–time float tunables; the thread converts them into the ISR's integer coefficients whenever they change. A fast 3 A step scores 2.25 in its first period; a 15–40 μs slew fires on the level, two thirds of the way down the dip the current–mode PI leaves on its own; ± 6 counts of noise on both samples score 0.6 (simulation: ± 8 counts false–trigger on the slope axis, so dV_a is set from the steady–state score of the candidate samples, which the panel shows). A feedback sample that jumps more than 256 counts in one period is held off for that cycle (two consecutive such samples are accepted), and a sample below a quarter of the reference for 500 μs trips the feedback–lost fault.

Model scale and dead time. V_{ref} and every voltage derived from the ADC use the *measured* scale, 793 μV per count (scope mean 792.6 mV at the 1000–count setpoint; an assumed 1 mV per count made every model voltage 26 % too large, so the pulse refilled a 26 % larger deficit with a 26 % too steep off–ramp and ended ∼1.3 A above the load). The inductor sees the CMP1 on–time minus the 50 ns rising dead time (272 ticks) in the estimator table, the current–loop inversion and the pulse solve.

8 Parameters

Symbol	Macro	Value
V_{in}	`BUCK_VIN_V`	6.0 V
V_{ref}	`BUCK_VOUT_MV`	0.793 V (measured)
L	`BUCK_L_H`	330 nH
C	`BUCK_C_F`	260 μF
r	`BUCK_R_MOHM`	114 mΩ
$t_{\min}$	`BUCK_OPT_TMIN_S`	100 ns
$T^{\max}_{\text{on}}$	`BUCK_OPT_TON_MAX_S`	1.5 μs ($< T_{\text{sw}} = 1.667\,\mu$s)
$T^{\max}_{\text{off}}$	`BUCK_OPT_TOFF_MAX_S`	6 μs
$I_{\text{pk,max}}$	`BUCK_OPT_IPK_MAX_A`	10 A
μ, Γ	`BUCK_OPT_GAMMA_*`	0.25, $[0.5, 2.0]$
μ_b, $\lvert b\rvert_{\max}$	`BUCK_OPT_IEQ_*`	0.5, 3 A
T_{tick}	T_{sw}/PERIOD	1.667 μs/8960